

Lesson 18

So far we have been using the joysticks and joystick buttons to do all of our controlling of the robot. In actual practice, the robot is controlled from a driver station which is a combination of joysticks, switches, and a SmartDashboard which is a virtual control panel displayed on the driver station laptop. The SmartDashboard is very powerful because it can have many tabs and many functions and only has to display the functions that are useful for the particular state of the system. There may be a tab for testing and another for calibration and another for trouble shooting. Usually the tab used during a competition would include among other functions the menu to select which Autonomous routine is being used as well as other unique functions. In this lesson we will learn how to set up this dashboard.

In the program you have been working with, look at RobotContainer.java and find the 'Setup SmartDashboard' section. This is where the Autonomous routine to be used is selected. You have already changed code here to select which autonomous methods are used, but you had to change the code to do it. With the SmartDashboard, you will do it before a match starts without having to change any code. This is where you will add other functions to activate commands or display parameters in real time.

The implementation we will be using is called Elastic. Elastic is automatically configured to look for a NetworkTables instance from the main robot. To connect to Elastic, go to the WPILib menu (Cntl+shift+P) and select 'Start Tool' and then 'Elastic'.

To start with a clean file, go to EDIT and Clear Layout and close any tabs if they exist.

Select Add Widget, Network Tables, SmartDashboard and then drag the SendableChooser to the grid. Close the Add Widget interface and save the layout. Now run autonomous and prove that you can use the Chooser to select the routine. You can add other autonomous commands and use the `m_chooser.addOption` to add them to the chooser widget.

A convenient switch to put on the SmartDashboard is to switch from Tank to Arcade drive. Do this by adding `SmartDashboard.putData("Tank Drive",`

`getTankDrivecommand()`); and similarly Arcade Drive in the Setup SmartDashboard options section in RobotContainer. Add these widgets to Elastic and test how easy it is to select Tank or Arcade Drive.

Add a command to turn the green LED on and another to turn it off. Add these to SmartDashboard and add the widgets to toggle it. Add another widget to indicate the green LED status. Hint: you can write to the dashboard directly inside the commands. The widget is binary and should indicate the color status. It is a good idea to save your layout after you test your changes.

The LED widgets should look something like this:

https://drive.google.com/file/d/10Bc3wSZY_XyLSvYRKKyvxxkxpk5Xp43OO/view?usp=drive_link

Add indicators for motor speeds.

At this point feel free to add other commands or displays on the dashboard. Save your layout.

Show off your good work to a mentor and let them know what you achieved and/or what you had trouble with.

Ask someone familiar with the competition robot to show you the driverstation and the dashboard with all the tabs that are used for the game.